

Part 3: Gradient Descent Manual calculation (Matrix form)

Given $\rightarrow$ The linear regression model is

$$\vec{y} = X_m + b$$

where

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

initial weights

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

initial bias

$$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

True outputs

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Learning rate

$$\alpha = 0.01$$

Number of samples

$$n = 2$$

* STEP 1:

Compute the predicted values

first calculate

$$X_m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\Rightarrow (1)(-1) + (3)(2) = \underline{5}$$

$$\Rightarrow (4)(-1) + (10)(2) = \underline{16}$$

$$X_m = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

$\rightarrow$ Add the bias vector

$$\vec{y} = X_m + b$$

$$= \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

* STEP 2:

COMPUTE THE ERROR

$$e = \hat{y} - y$$

$$= \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

* STEP 3:

MEAN SQUARED ERROR

The MSE cost function is

$$J(m, b) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

Since given %

$$n = 2$$

$$J = \frac{1}{2} (1^2 + 11^2)$$

$$= \frac{1}{2} (122)$$

$$J = 61$$

* STEP 4:

DERIVE THE GRADIENTS.

The cost function is

$$J(m, b) = \frac{1}{n} (\hat{y} - y)^T (\hat{y} - y)$$

Since

$$\hat{y} = X_m + b$$

* Substitute into the cost function

$$J = \frac{1}{n} (X_m + b - y)^T (X_m + b - y)$$

* Taking the derivative with respect to the weight vector.

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T (\hat{y} - y)$$

* Taking the derivative with respect to the bias vector.

$$\frac{\partial J}{\partial b} = \frac{2}{n} (\hat{y} - y)$$

Since

$$n = 2$$

$$\nabla_m = X^T (\hat{y} - y)$$

$$\nabla_b = (\hat{y} - y)$$

* ITERATION 1

Gradient for m

Transpose for X

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

multiply

$$\nabla_m = X^T e$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= 1(1) + 4(11) = 45$$

$$= 3(1) + 10(11) = \underline{\underline{113}}$$

Therefore

$$\nabla_m = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

Gradient of bias

$$\nabla_b = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Update m

$$m_1 = m - \alpha \nabla_m$$

$$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \underline{\underline{\begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}}}$$

Update b

$$b_1 = b - \alpha \nabla_b$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

* ITERATION 2:

Using

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

Prediction

$$X_m = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

Adding bias

$$\hat{y} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

Error

$$e = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

Cost

$$\begin{aligned} J &= \frac{1}{2} \left((-2.85)^2 + (-2.21)^2 \right) \\ &= \underline{\underline{6.5033}} \end{aligned}$$

GRADIENT

$$\nabla_m = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

Gradient for bias

$$\nabla_b = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

Update

$$m = \begin{bmatrix} -1.3331 \\ 9.1765 \end{bmatrix}$$

$$b = \begin{bmatrix} 1.0135 \\ 0.9121 \end{bmatrix}$$

ITERATION 3:

Prediction

$$\vec{y} = \begin{bmatrix} 3.2149 \\ 7.3465 \end{bmatrix}$$

Error

$$e = \begin{bmatrix} -1.7851 \\ 1.3465 \end{bmatrix}$$

Cost

$$J \approx 2.50$$

Gradient

$$\nabla_m = \begin{bmatrix} 3.6009 \\ 8.1087 \end{bmatrix}$$

Gradient for bias

$$\nabla_b = \begin{bmatrix} -1.7851 \\ 1.3465 \end{bmatrix}$$

update

$$m = \begin{bmatrix} -1.3691 \\ 0.8986 \end{bmatrix}$$

$$b = \begin{bmatrix} 1.0364 \\ 0.8986 \end{bmatrix}$$

SUMMARY TABLE

Iteration	m	b	MSE
Initial	$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	61.0000
1	$\begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$	$\begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$	6.5033
2	$\begin{bmatrix} -1.3334 \\ 1.1765 \end{bmatrix}$	$\begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$	2.5033
3	$\begin{bmatrix} -1.3691 \\ 1.0954 \end{bmatrix}$	$\begin{bmatrix} 1.0364 \\ 0.8986 \end{bmatrix}$	≈ 2.13

Observation

The values of the weight vector m and bias vector b are updated in the direction opposite to the gradient at each iteration. This progressively reduces the Mean Squared Error (MSE), which decreases from 61.00 initially to approximately 2.13 after the third update. This shows that gradient descent is successfully minimizing the cost function and improving the model's predictions.